RF-Based Drone Identification Using Machine Learning

[23ET7PE3ML]

Dept. of Electronics and Telecommunication Engineering

BMS College of Engineering, Bengaluru

Guided by

Dr. C Gururaj
Professor, ECE

Presented by

Utkarsh Sinha (1BM23ET055)
Vishal Joshy (1BM23ET058)
Yash Gupta (1BM23ET061)

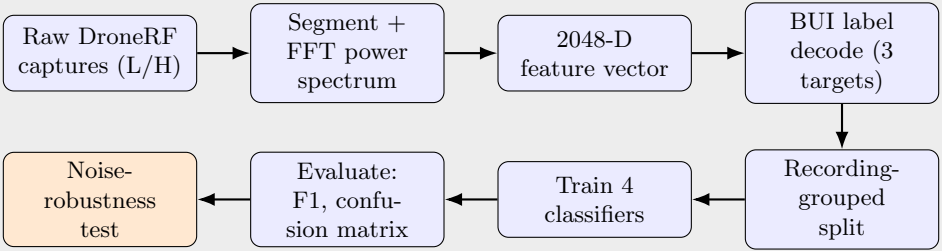
Introduction

- Drones are cheap and everywhere; their misuse raises privacy and security risks near airports and public spaces.
- Cameras and radar need line of sight and good conditions. Drones instead keep talking to their controllers over RF links in the 2.4 GHz band.
- Listening to those RF signals detects a drone passively, without its cooperation and without line of sight.
- Knowing a drone is present is not enough; a useful system also says *which* drone and *what* it is doing.
- This project classifies recorded DroneRF signals for three tasks: presence, family, and operating condition.

Problem Statement

- A single sensor cannot read drone presence reliably in all conditions, yet security and airspace management depend on it.
- The 2.4 GHz band is shared with Wi-Fi and Bluetooth, so recorded RF signals are noisy.
- Segments cut from one recording are nearly identical. If they fall in both train and test sets, accuracy looks better than it is (data leakage).
- Goal: classify presence, family, and operating condition from spectral features, under an evaluation that explicitly avoids leakage.

Proposed Workflow



Dataset Used: DroneRF

Class (Family)	**BUI codes**	**Modes**	**Role**
Background (no drone)	00000	1	Negative class
Parrot Bebop	10000–10011	4	Drone family 1
Parrot AR	10100–10111	4	Drone family 2
DJI Phantom	11000	1	Drone family 3

- 227 recording pairs (low `L` + high `H` sub-bands) in the 2.4 GHz ISM band.
- Re-segmenting yields **22,700** feature vectors for training.

Label Decoding from the Filename

- Every file name starts with a 5-digit **BUI** code, e.g. `10111H_3.csv`.
- All three labels are decoded from this one code, so no manual labelling is needed.

Target	**Rule**
presence (2 classes)	0 if background, else 1
family (4 classes)	background / Bebop / AR / Phantom
mode (10 classes)	one index per BUI code

Feature Extraction

- Each capture is cut into segments of $L = 100{,}000$ samples.
- For each segment the mean is removed and a 2048-point FFT magnitude is taken, keeping the upper half (1024 bins):

$$X_f = \left| \mathrm{FFT}_M(x - \bar{x}) \right|, \qquad f = \tfrac{M}{2} + 1, \ldots, M.$$

- Low and high sub-bands are joined and squared to give a power spectrum.
- Result: a **2048-dimensional** feature vector per segment.

Literature Survey (1)

Paper	**Method**	**Why it matters to us**
Al-Sa'd et al., 2019 (FGCS)	Deep neural networks on DroneRF	Created our dataset and the three-task structure we follow.
Allahham et al., 2019 (Data in Brief)	DroneRF data descriptor	Documents the file layout and BUI labelling we decode.
Al-Emadi et al., 2020	XGBoost, 3 tasks	Closest prior work; confirms XGBoost is strong and that mode is the hard task.
Ezuma et al., 2019 (Aerospace)	kNN on RF fingerprints	Classic classical-ML baseline using statistical RF features.
Ezuma et al., 2020 (OJCOMS)	ML under Wi-Fi/BT interference	Motivates our noise-robustness test in the shared band.

Literature Survey (2)

Paper	**Method**	**Why it matters to us**
Medaiyese et al., 2022 (PMC)	Wavelet features + ML	The main alternative to our FFT/PSD features.
LUMEN, 2026 (Sensors)	PSD features, lightweight net	Uses the same power-spectral features we chose, on edge devices.
Robustness study, 2024 (Sensors)	RF UAV detector robustness	Frames our SNR degradation test as standard practice.
Ensemble FP, 2024 (Sensors)	Ensemble ML under jamming	Supports our Random Forest / XGBoost choice under interference.
Complex-valued CNN, 2026 (Sensors)	Low-SNR recognition	Recent baseline on a DroneRF-family dataset at low SNR.

Research gap: most work is deep-learning or omits a leakage check. We compare four interpretable classical models across all three tasks under a recording-grouped split.

Algorithms Used and Why

Algorithm	**Why included**	**Behaviour on this data**
Logistic Regression	Simple linear baseline; if it wins, the task is easy	Perfect on presence/family, weak (0.375) on mode
Random Forest	Bagging ensemble; handles non-linear patterns	Stable, slightly weaker on family (0.898)
SVM (RBF kernel)	Curved class boundaries	Strong on easy tasks; limited on mode (5k-sample cap)
XGBoost	Gradient-boosted trees; usually the strongest	**Best overall**, and clearly best on the hard mode task

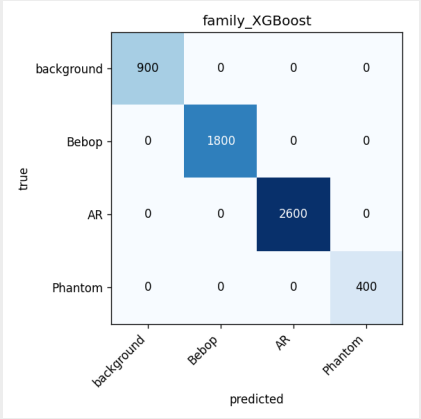
Comparison takeaway: all models solve the easy tasks, so they look the same there. XGBoost separates from the rest only on the hard 10-class mode task, which is where the comparison actually matters.

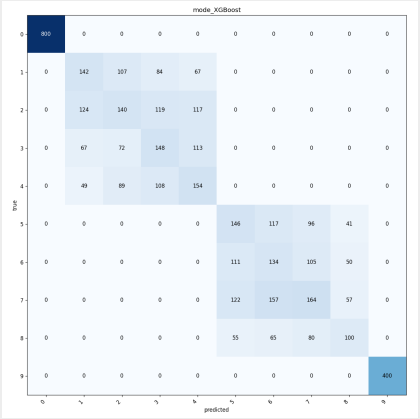
Results: Macro-F1 by Model and Task

Task	**LR**	**RF**	**SVM**	**XGBoost**
Presence	1.000	0.978	1.000	1.000
Family	0.999	0.898	0.977	**1.000**
Mode (5-fold)	0.375	0.378	0.160	**0.492**

- Presence and family are essentially solved.
- The 10-class mode task is hard; XGBoost leads clearly at 0.492 ($\pm$0.009 over 5 folds).

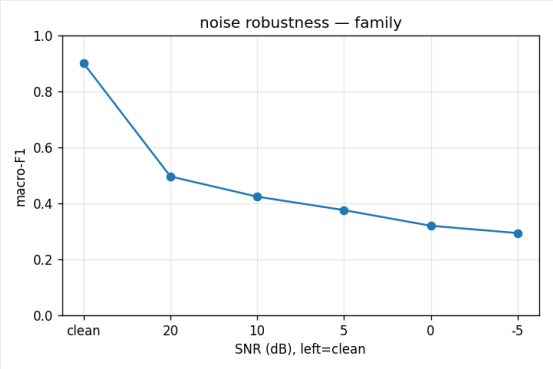
Confusion Matrices (XGBoost)





Family (left) is a clean diagonal. Mode (right) shows where the 10 operating conditions get confused.

Noise Robustness (Family)



SNR	Macro-F1
clean	0.900
20 dB	0.496
10 dB	0.424
5 dB	0.376
0 dB	0.320
−5 dB	0.294

Accuracy roughly halves by 20 dB SNR: spectral features alone are not robust to heavy interference.

Advantages

- Passive detection: works without line of sight or good lighting.
- Interpretable, low-cost classical models; fast to train.
- Leakage-aware evaluation gives honest, trustworthy numbers.
- Single pipeline answers all three questions (presence, family, mode).
- Noise test quantifies real-world robustness instead of assuming it.

Limitations

- Operating-condition (mode) classification remains hard (best F1 0.492).
- Spectral features degrade quickly under heavy noise and interference.
- Only three drone families in the dataset; limited model diversity.
- SVM does not scale, so it used a 5000-sample training subset.
- Dataset-based study; no live hardware or real-world deployment.

Applications

- Passive drone surveillance around airports and restricted areas.
- Early warning for unauthorised drones at public events.
- Low-cost monitoring on software-defined radios.
- Identifying drone type to assess the level of threat.
- A baseline for larger RF drone-classification research.

Conclusion

- Built a classical ML pipeline for RF drone identification on DroneRF.
- Compared four models across presence, family, and mode under a leakage-aware split.
- Presence and family are reliably classified; XGBoost is best on the hard mode task at 0.492.
- A noise test measured how fast spectral features degrade under interference.
- Interpretable, low-cost models stay competitive with deep learning on this dataset.

References

1. M. F. Al-Sa'd et al., "RF-based drone detection and identification using deep learning approaches," *Future Generation Computer Systems*, vol. 100, 2019.
2. M. S. Allahham et al., "DroneRF dataset," *Data in Brief*, vol. 26, 2019.
3. S. Al-Emadi, F. Al-Senaid, "Drone detection using machine learning," *Proc. IEEE IIT*, 2020.
4. M. Ezuma et al., "Micro-UAV detection from RF fingerprints," *Proc. IEEE Aerospace Conf.*, 2019.
5. M. Ezuma et al., "Detection and classification of UAVs in the presence of Wi-Fi and Bluetooth interference," *IEEE OJ-COMS*, vol. 1, 2020.
6. O. O. Medaiyese et al., "Wavelet transform analytics for RF-based UAV detection," *Pervasive and Mobile Computing*, vol. 82, 2022.
7. "Robustness of deep-learning-based RF UAV detectors," *Sensors*, vol. 24(22), 2024.
8. "Composite ensemble learning for passive drone RF fingerprinting," *Sensors*, vol. 24(17), 2024.
9. "RF signal recognition of UAV using complex-valued CNN," *Sensors*, vol. 26(2), 2026.
10. "LUMEN: lightweight UAV network for PSD-based RF fingerprinting," *Sensors*, vol. 26(10), 2026.